

MATH GU4053 INTRO TO ALGEBRAIC TOPOLOGY
HOMEWORK 7

DUE: WEDNESDAY, MARCH 25, 2026, **BEFORE 23:59PM**

SUBMIT THE COMPLETED ASSIGNMENT TO <https://www.gradescope.com>

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- This assignment has **5 questions** on **two pages**.
 - This assignment counts towards 2% of your final grade (up to a maximum of 10% for all homework).
 - While collaborative work is allowed and generally beneficial, you must write up the solutions *on your own* and *in your own words*.
 - Rigor, clarity, and proper use of mathematical notation count! The work you submit for evaluation will require showing all steps and a clear explanation of what you have done. The explanations need not be lengthy to be clear.
 - Solutions based on an advanced result which has not been covered in this class, or one of its prerequisites, will not receive full credit. (Especially if the use of an advanced result significantly changes the difficulty of the problem.)
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1. Let

$$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$$

be a short exact sequence, and assume that there exists a homomorphism $s : C \rightarrow B$ such that $g \circ s = \text{id}$. Prove that the homomorphism

$$\begin{aligned} A \oplus C &\rightarrow B \\ (a, c) &\mapsto f(a) + s(c) \end{aligned}$$

is an isomorphism.

2. (a) Define homomorphisms $f : \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$ and $g : \mathbb{Z}/4\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ such that

$$0 \rightarrow \mathbb{Z}/2\mathbb{Z} \xrightarrow{f} \mathbb{Z}/4\mathbb{Z} \xrightarrow{g} \mathbb{Z}/2\mathbb{Z} \rightarrow 0$$

is a short exact sequence. *Justify your answer.*

(b) Prove that there does not exist a homomorphism $s : \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$ such that $g \circ s = \text{id}$.

(c) Prove that there exists a function $s : \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$ such that $g \circ s = \text{id}$.

3. The *reduced singular chain complex* $\tilde{C}_*(X)$ of a topological space X is defined by

$$\tilde{C}_p(X) = \begin{cases} C_p(X) & \text{for } p \geq 0, \\ \mathbb{Z} & \text{for } p = -1, \\ 0 & \text{for } p \leq -2, \end{cases}$$

and $\partial : \tilde{C}_p(X) \rightarrow \tilde{C}_{p-1}(X)$ is defined as the singular boundary $C_p(X) \rightarrow C_{p-1}(X)$ for $p > 0$, while

$$C_0(X) = \tilde{C}_0(X) \xrightarrow{\partial} \tilde{C}_{-1}(X) = \mathbb{Z}$$

is defined by sending all $\sigma : \Delta^0 \rightarrow X$ to $1 \in \mathbb{Z}$.

- (a) Prove that $(\tilde{C}_*(X), \partial)$ is a chain complex.
- (b) Let $A_{-1} = \mathbb{Z}$ and $A_p = 0$ for $p \neq -1$, and define $\partial : A_p \rightarrow A_{p-1}$ as the zero homomorphism for all p . Explain how to define a short exact sequence of chain complexes of the form

$$0 \rightarrow A_* \rightarrow \tilde{C}_*(X) \rightarrow C_*(X) \rightarrow 0.$$

4. Let X be a topological space, let $\tilde{C}_*(X)$ be the chain complex defined in the previous exercise, and let $\tilde{H}_p(X) = H_p(\tilde{C}_*(X), \partial)$ be its homology groups, $p \in \mathbb{Z}$.

According to a theorem proved in class (notes, theorem 6.5.5) the short exact sequence in the previous exercise induces a long exact sequence of homology groups. In this problem will investigate

- (a) Briefly explain why $H_p(A_*, \partial) \cong A_p$ for all $p \in \mathbb{Z}$, and deduce exact sequences of the form

$$0 \rightarrow \tilde{H}_p(X) \rightarrow H_p(X) \rightarrow 0$$

for all $p \in \mathbb{Z} \setminus \{0, -1\}$.

- (b) Deduce an exact sequence of the form

$$0 \rightarrow \tilde{H}_0(X) \rightarrow H_0(X) \xrightarrow{\delta} \mathbb{Z} \rightarrow \tilde{H}_{-1}(X) \rightarrow 0.$$

- (c) Use the definition of the connecting homomorphism (notes, Lemma 6.5.4) to evaluate $\delta([x]) \in \mathbb{Z}$ for any $x : \Delta^0 \rightarrow X$. Deduce a formula for $\tilde{H}_{-1}(X)$.

The formula for $\tilde{H}_{-1}(X)$ will have two cases, depending on whether $X = \emptyset$ or $X \neq \emptyset$. In the case $X = \emptyset$, you may use that we already calculated $H_0(\emptyset) \cong 0$.

5. Let X be a topological space, let $x_0, x_1 \in X$ be two distinct points, and let $A = \{x_0, x_1\}$. Part of the long exact sequence for the pair (X, A) looks like

$$\cdots \rightarrow H_1(X, A) \xrightarrow{\delta} H_0(A) \rightarrow \cdots$$

In this exercise, it may be used without proof that $H_0(A) \cong \mathbb{Z}^2$, with basis elements $[x_0]$ and $[x_1] \in H_0(A)$.

- (a) Prove that if X is path connected, then the connecting homomorphism $\delta : H_1(X, A) \rightarrow H_0(A)$ is non-zero.
- (b) Now assume $X = \mathbb{R}$, $x_0 = 0$, and $x_1 = 1$. Describe an explicit chain $c \in C_1(X, A)$ with $\partial c = 0$ and $\delta([c]) \neq 0 \in H_0(A)$.